

Ex.No.5: MIMO System Simulation with BER Analysis

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

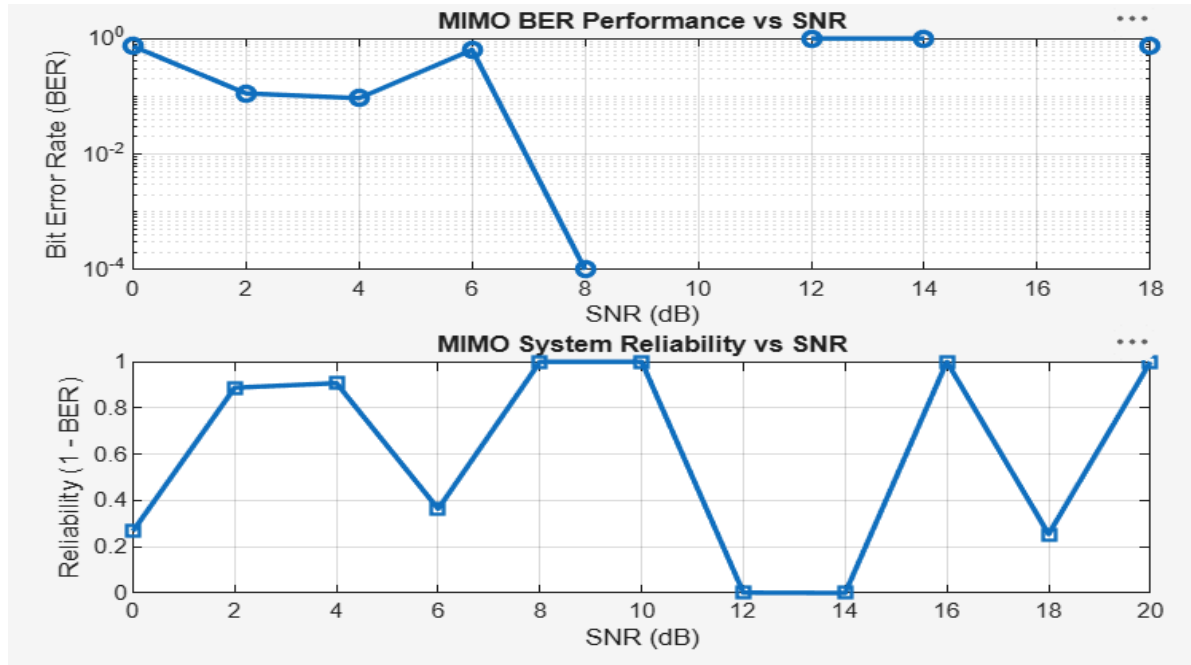
Objective:

1. To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.
2. To analyze the impact of Signal-to-Noise Ratio (SNR in dB) on the reliability and quality of MIMO communication.
3. To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;
% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));
for i = 1:length(SNR)
    bits = randi([0 1],10000,1);
    symbols = 2*bits-1; % BPSK
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
    tx = H(1,1)*symbols' + noise(1,:);
    rx = real(tx) > 0;
    BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
% ----- Graph 1: BER vs SNR -----
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% ----- Graph 2: SNR vs Quality Indicator -----
subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on
```

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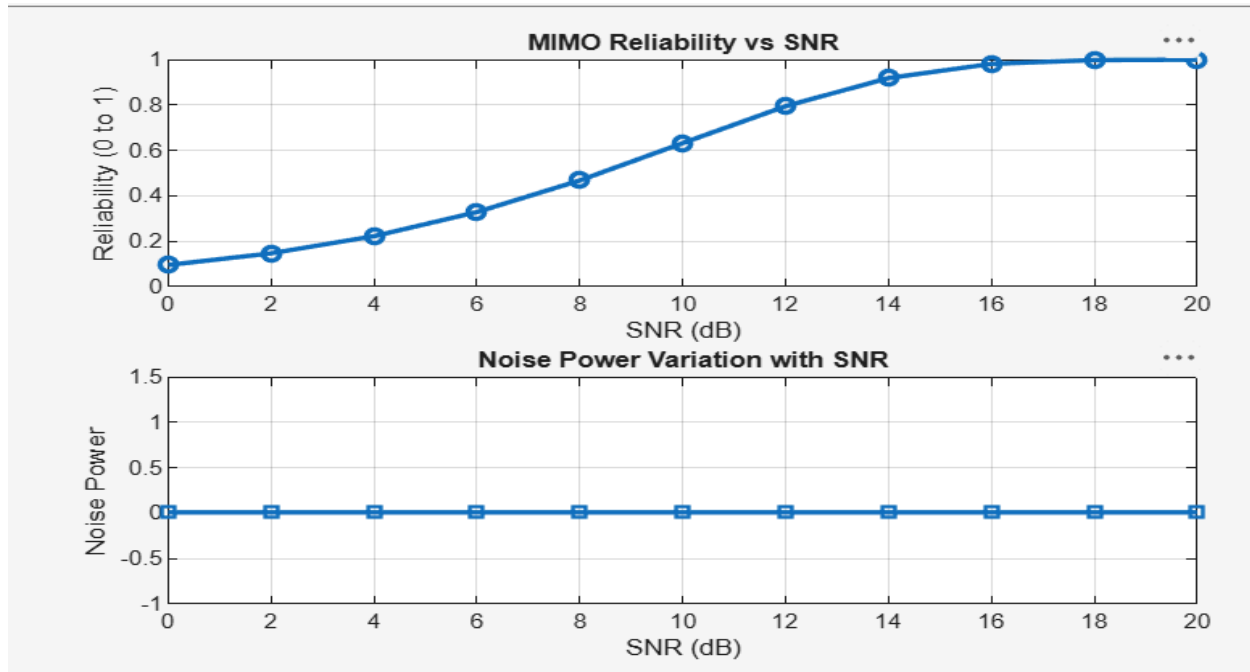


Objective 2 Matlab Code and Output:

```
clc; clear; close all;
% SNR range (dB)
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
% Simulated reliability metric (inverse BER model)
Reliability = zeros(1,length(SNR));
for i = 1:length(SNR)
    % simple channel model effect
    H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
    noise_power = 1/SNR(i);
    % reliability increases with SNR
    Reliability(i) = 1 - exp(-SNR(i)/10);
end
figure
% ----- Graph 1: SNR vs Reliability -----
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
```

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```
% ----- Graph 2: SNR vs Noise Effect -----  
subplot(2,1,2)  
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)  
title('Noise Power Variation with SNR')  
xlabel('SNR (dB)')  
ylabel('Noise Power')  
grid on
```



Objective 3 Matlab Code and Output:

```
clc; clear; close all;  
% Resource Block parameters  
rb_bw = 180e3; % 180 kHz per RB  
% Users  
users = 1:5;  
% Allocated Resource Blocks per user  
alloc_RB = [10 12 8 15 5];  
% Modulation efficiency (bits/symbol)  
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.  
% Throughput calculation (Mbps)  
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;  
% Display results  
disp('User Throughput (Mbps):')  
disp(throughput)
```

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figure

% ----- Graph 1: Throughput per User -----

subplot(2,1,1)

plot(users, throughput, '-o','LineWidth',2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2: RB vs Throughput -----

subplot(2,1,2)

plot(alloc_RB, throughput, '-s','LineWidth',2)

title('Resource Blocks vs Throughput')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on

